TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

AACS2204 OBJECT-ORIENTED PROGRAMMING TECHNIQUES

ASSIGNMENT

SEMESTER 202501

Programme (Year & Group)	:	DCS1S3
Tutor Name	:	Joan Hau
Tutorial Group	:	1
Date Submitted	:	2/5/2025

No	Name	Student ID	Marks			Total Marks
			Section A	Section B	Section C	
1	NG KIAH QUAN	2400429	23	45	22	90
2	KITT LEE JIA LING	2401738	23	45	22	90
3	WONG SUN HAN	2401927	0	45	22	67
4						
5						

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

AACS2204 OBJECT-ORIENTED PROGRAMMING TECHNIQUES

Diploma in Computer Science (DCS) Year 1 Semester 3

Tutorial Group: Group 1

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3	WONG SUN HAN	2401927	HAN	2/27/2025
4				



FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

AACS2204 OBJECT-ORIENTED PROGRAMMING TECHNIQUES

Diploma in Computer Science (DCS)

Year 1 Semester 3

Tutorial Group: Group 1

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I, Name NG KIAH QUAN Student ID 2400429

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Signature : *Kiahquan*

Date : 2/27/2025

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AACS2204 OBJECT-ORIENTED PROGRAMMING TECHNIQUES

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AACS2204 OBJECT-ORIENTED PROGRAMMING TECHNIQUES

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Year 1 Semester 3

Tutorial Group: Group 1

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Assignment Assessment Rubrics

No	Student Name	Student ID	Final Marks
1	NG KIAH QUAN	2400429	90
2	KITT LEE JIA LING	2401738	90
3	WONG SUN HAN	2401927	67
4			

Programme of Study	D (CS1)
Tutorial Group	Group (1)

CLO 2	Demonstrate an object-oriented program using appropriate programming fundamentals with regards to arrays, methods and exception handling. (P4, PLO3)

Section A: Program Output & Correctness - Individual Marks (25 Marks)

Task	Weight	Max. Marks	Excellent	Good	Average	Poor	Very Poor	Scores (Point x Weight)	Remarks
			5	4	3	2	1		
Completeness of the program	**2.0**	**10**	Completed 100% of the functional requirements. All operations are implemented correctly.	Completed all of the functional requirements but implemented with minor errors.	Completed most of the functional requirements but implemented with errors.	Completed < 50% of the functional requirement and < 50% of the operations implemented correctly.	Did not complete any of the functional requirement at all		

No	Marks
1	10
2	10
3	
4	

Exception Handling	**1.0**	**5**	Output is correct with comprehensive exception handling.	Output is correct with appropriate exception handling.	Output is correct with little exception handling.	Output is somewhat correct with little proper exception handling.	Output is totally wrong with no proper exception handling.	**No** / **Marks** 1 / 4 2 / 4 3 / 4 /	no date validation
Design of the Output	**1.0**	**5**	Excellent formatted output. Program displays more than expected	Good formatted output and program displays as expected.	Appropriate output design and display as expected.	Poorly designed output.	Minimum to no formatted output with messy display.	**No** / **Marks** 1 / 4 2 / 4 3 / 4 /	
Program Code Quality & Standard	**1.0**	**5**	All identifiers strictly conform to the standard Java naming convention and all are meaningful	Most of the identifiers conform to the standard Java naming convention and are meaningful	Some of the identifiers conform to the standard Java naming convention and some are meaningful	Limited numbers of the identifiers conform to the standard Java naming convention or are barely meaningful.	All of the identifiers conform to the standard Java naming convention or not meaningful	**No** / **Marks** 1 / 5 2 / 3 / 4 /	
						Total Marks (25 marks):	**1**	23	

	2	23
	3	0
	4	

CLO 3	Analyse the concepts of encapsulation, inheritance and polymorphism based on programming problems. (C4, PLO2)								
Section B: Object-Oriented Concept - Group Marks (45 Marks)									
Task	**Weight**	**Max. Marks**	**Excellent**	**Good**	**Average**	**Poor**	**Very Poor**	**Scores (Point x Weight)**	**Remarks**
			5	**4**	**3**	**2**	**1**		
Class Implementation	**1.0**	**5m**	Identified the proper classes, methods and attributes to solve particular problems.	Identified appropriate classes, methods and attributes to solve particular problems with minor errors.	Able to identify only some classes, methods and attributes to solve particular problems.	Inappropriate classes, methods and attributes are used.	Unable to show understanding on the usage of classes, methods and attributes.	5	
Object Implementation	**1.0**	**5m**	Able to create a structure of objects collaborating among themselves to carry out tasks properly.	Able to create a structure of object collaborating among themselves to carried out task properly with little minor mistakes	Structure of collaborating object created with some mistakes	Structure of collaborating objects created is barely correct.	Wrong structure of object collaboration like one object doing everything itself.	5	
Abstraction	**1.0**	**5m**	Define the class at the proper level of abstraction (abstract classes and methods).	Define the class at the proper level of abstraction with minor mistake	Classes defined at the proper level of abstractions with some mistakes	< 50% of correct abstraction is used	Lack of abstraction (abstract classes and methods).	5	

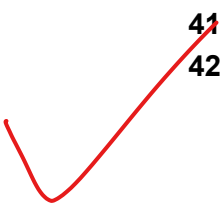
Encapsulation	**2.0**	**10m**	Completely correct implementation of encapsulation (private modifier, setter and getter methods).	Correct implementation of encapsulation with only some minor mistakes	Acceptable amount of correct implementation of encapsulation is applied in program	< 50% of correct implementation for encapsulation applied in program	Completely incorrect implementation of encapsulation (private modifier, setter and getter methods)	10	
Inheritance	**2.0**	**10m**	Completely correct use of inheritance (correct use of extends keyword, super & sub class)	Correct implementation of inheritance with only some minor mistakes	Acceptable amount of correct implementation of inheritance is applied in program	< 50% of correct implementation for inheritance applied in program	Completely incorrect use and implementation of inheritance	10	
Polymorphism	**2.0**	**10m**	Completely correct use and implementation of polymorphism. Methods toString() & equals() correctly overridden.	Correct implementation of polymorphism with only some minor mistakes	Acceptable amount of correct implementation of polymorphism is applied in program	< 50% of correct implementation for polymorphism applied in program	Completely incorrect use and implementation of polymorphism.	10	
							Total Marks (45 marks):	45	

CLO 3	Analyse the concepts of encapsulation, inheritance and polymorphism based on programming problems. (C4, PLO2)								
Section C: Object-Oriented Concept - Group Marks (30 Marks)									
Task	**Weight**	**Max. Marks**	**Excellent**	**Good**	**Average**	**Poor**	**Very Poor**	**Scores (Point x Weight)**	**Remarks**
			5	**4**	**3**	**2**	**1**		
Cohesion	**1.0**	**5m**	The code demonstrates exceptional cohesion with all modules, classes, or functions having a clear and single responsibility.	The code demonstrates good cohesion overall where most modules, classes, or functions have a clear and single responsibility	Some modules, classes, or functions have a clear and single responsibility, but others may be slightly ambiguous	Few modules, classes, or functions have a clear and single responsibility.	Modules, classes represent more than one entity and lack of clear responsibility. Poor class cohesion.	5	
Coupling	**1.0**	**5m**	Minimal or no direct dependency between modules, classes, or functions. Coupling is effectively managed, resulting in highly maintainable and modular code	Most modules, classes, or functions have minimal direct dependencies.	Some modules, classes, or functions have moderate direct dependencies.	Many modules, classes, or functions have significant direct dependencies.	Modules, classes, or functions have excessive and tightly coupled dependencies which make module be harder to reuse or test.	5	
Association, Aggregation	**2.0**	**10m**	Associations between classes	Associations between classes	Some associations	Several associations	Completely incorrect use and		

and Composition			are accurately identified and properly established.	are generally identified and properly established.	between classes may be missing or improperly established.	between classes are missing or improperly established.	implementation of association, aggregation and composition relationship.	6	Unable to show the relationship
UML Class Diagram	**2.0**	**10m**	The UML class diagram demonstrates excellent understanding and implementation. All classes, attributes, and methods are accurately represented with appropriate visibility, data types, and associations.	Most classes, attributes, and methods are accurately represented with appropriate visibility, data types, and associations.	Some classes, attributes, and methods may be missing or inaccurately represented, with visibility, data types, or associations not fully defined.	Many classes, attributes, and methods are missing or inaccurately represented, with visibility, data types, or associations not adequately defined.	Classes, attributes, and methods are missing or inaccurately represented, with visibility, data types, or associations lacking clarity or definition.	6	donot reflect what is done or code.
							Total Marks (30 marks):	22	

Table of Contents

1.0 Introduction **14**
1.1 User Management 14
1.2 Movie Management 14
1.3 Booking Management 14
2.0 Task Allocation **15**
3.0 UML Diagram **16**
4.0 System Workflows **17**
4.1 Main Program 17
4.1.1 Home 17
4.1.2 Staff Login 18
4.1.3 Movie Management 19
4.1.4 Showtime Management 25
4.1.5 Hall Management 29
4.1.6 Register Customer 31
4.1.7 Customer Login 33
4.1.8 Booking Management 38
5.0 Conclusion **41**
6.0 References **42**



1.0 Introduction (Cinema Ticketing System)

The Cinema TIcketing System (CTS) is a software solution designed to streamline the processes of ticket booking, movie management and user account management. As the entertainment industry continues to grow, the need for efficient ticket management becomes increasingly important. This system aims to optimize the process of movie ticket booking for both the customers and staff, ensuring seamless and smooth cinema experiences for our customers, smoother working operations, and efficiently managing Movies, Showtimes, Customers, Ticket Booking, payments, and more. By integrating various functionalities into a cohesive platform, the CTS enhances the booking experience, reduces errors, improves user experience, and increases efficiency and effectiveness of staff operations.

1.1 User Management:

The User Management System module is an essential component of the CTS that facilitates the interaction between users and the system. For customers, the module allows them to register themselves as a CTS member, enabling them to log in to their accounts and access key features such as booking movie tickets and viewing showtimes, and even updating their account details or deleting their account. Meanwhile, staff members can use the same module to securely log into the system and grant them access to all administrative and operational features of the CTS. This functionality ensures seamless account management for both customers and staff, contributing to the overall efficiency of the CTS.

1.2 Movie Management

The Movie Management System module serves as the backbone of the CTS, because a CTS without a movie cannot do anything. On the user's side, it enables users to effectively view the movie listings of the available movies, search the movie for viewing more specific details of the movies, such as title, genre, duration, director, language, rating, and cast information. On the staff side, it enables staff to add, view, update, delete, and search movies. In addition, Movie Management also includes Showtime Management and Hall Management. It also enables staff to add, view, and delete the showtime and hall. This comprehensive management ensures that the staff and users can modify and view reliable movie information.

1.3 Booking Management

The Booking Management System module is responsible for the process of booking a ticket for a movie. This module includes several essential functions, including enabling users to choose their preferred showtime and movie, view the seat layout for available seats, confirm booking details, payment, and print tickets for the users. In addition, users can view all the showtime listings based on the date, which is more convenient for the users. With the clear seat layout, if the number of seats is empty, meaning that the seats are available, if the number of seats is “X”, meaning that the seats are unavailable. This provides simple instructions to the users on which seats are available. All the selections are made by the users, It will ask users if they want to proceed payment or not, if users proceed with the payment and succeed, it will produce payment successful information and generate a ticket for the users.

2.0 Task Allocation

Student Name	Modules Handled	Description
NG KIAH QUAN	Movie Management, Booking Management	Implement CRUD for movies, CRD for showtimes and hall. Implement booking processes such as showtime selection, seat layout, seat selection, proceed payment and ticket.
KITT LEE JIA LING	UserManagement	Implement CRUD for User registration, User profile editing, User account deletion, user login, and user logout
WONG SUN HAN	-	-



3.0 UML Diagram

FileManager

+ saveToFile(fileName: String, object: T) : void
+ loadFromFile(fileName: String): T

ShowtimeController

- showtimes: List<Showtime>
- FILE_NAME: String
- scanner: Scanner
- hallController: HallController
- movieController: MovieController

+ ShowtimeController(hallController: HallController, movieController: MovieController)
+ addShowtime(): void
+ viewShowtimes(): void
+ saveShowtimes(): void
+ deleteShowtime(): void
- isShowtimeConflict(dateTime: LocalDateTime, hall: Hall): boolean
+ getShowtimes(): List<Showtime>
+ getShowtimeById(id: String): Showtime
+ run(): void

IMAXHall

- soundSystemBrand: String
- BASE_PRICE: double

+ IMAXHall(id: String, hallName: String, soundSystemBrand: String)
+ getSoundSystemBrand(): String
+ setSoundSystemBrand(soundSystemBrand: String): void
+ getBasePrice(): double
+ calculatePrice(): double
+ toString(): String

NormalHall

- hasReclinerSeats: boolean
- BASE_PRICE: double

+ NormalHall(id: String, hallName: String, hasReclinerSeats: boolean)
+ hasReclinerSeats(): boolean
+ getBasePrice(): double
+ calculatePrice(): double
+ toString(): String

MovieController

- movies: List<Movie>
- FILE_NAME: String
- scanner: Scanner

+ MovieController()
+ addMovie(): void
+ viewMovie(): void
+ updateMovie(): void
+ deleteMovie(): void
+ searchMovie(): void
+ run(): void
+ getMovieById(id: String): Movie
- saveMovie(): void
- loadMovie(): void

Movie

- id : String
- title : String
- genre : String
- runningTime : String
- subtitles : String
- director : String
- casts : String
- spokenLanguage : String
- description : String
- classifications : String

+ Movie(id : String, title: String, genre: String, runningTime: String, subtitles: String, director: String, casts: String, spokenLanguage: String, description: String, classifications: String)
+ Movie()
+ accessors and mutators
+ toString() : String

HallController

- halls: List<Hall>
- FILE_NAME: String
- scanner: Scanner

+ HallController()
+ addHall(): void
+ viewHalls(): void
+ deleteHall(): void
+ saveHalls(): void
+ getHallById(id: String): Hall
+ run(): void

Hall

- id : String
- hallName : String

+ Hall()
+ Hall(id: String, hallName: String)
+ getId() : String
+ setId(id: String) : void
+ getHallName() : String
+ setHallName(hallName String) : void
+ *getBasePrice(): double*
+ *calculatePrice(): double*
+ toString(): String

ThreeDHall

- provides3DGlasses: boolean
- BASE_PRICE: double

+ ThreeDHall(id: String, hallName: String, provides3DGlasses: boolean)
+ isProviding3DGlasses(): boolean
+ getBasePrice(): double
+ calculatePrice(): double
+ toString(): String

MainMenu

+ main(args: String[]): void
+ displayMenu() : void
+ displayGoodbye(): void

Showtime

- id : String
- dateTime : LocalDateTime
- hall : Hall
- movie : Movie

+ Showtime()
+ Showtime(showtimeId: String, dateTime: String, hall: Hall, movie: Movie)
+ getId() : String
+ getDateTime(): LocalDateTime
+ getHall(): Hall
+ getMovie(): Movie
+ setDateTime(dateTime: LocalDateTime) : void
+ toString() : String

Seat

+ Seat()
+ Seat(id: String)
+ toString() : String

StaffMenu

+ show(staff: Staff): void

CustomerMenu

+ show(customer : Customer): void
+ customerExist (customer : Customer) : boolean

Item

id: String
isBooked: boolean

+ Item(id: String)
+ getId(): String
+ setId(id: String): void
+ isBooked(): boolean
+ setBooked(isBooked: boolean): void
+ toString() : String
+ generateRandomId(prefix: String): String

RegisterCust

- customerList: ArrayList<Customer>
- FILE_NAME: String

+ register(scan : Scanner): void
- isUserIDTaken(userID: String): boolean
+ verifyCust(password : String, password2 : String) : boolean
+ confirmCust(name : String, userID : String, telNo : int, email : String, scan : Scanner) : int
- saveCustomers(): void
- loadCustomer(): void

Booking

- showtime: Showtime
- seatId: Seat

+ Booking()
+ Booking(bookingId: String, showtime: Showtime, seatId: Seat)
+ getShowtime(): Showtime
+ getSeatId(): Seat
+ toString() : String

Customer

- telNo : int
- userID : String
- email : String

+ Customer(name: String, password: String, telNo : int, userID: String, email: String)
+ getTelNo() : int
+ getUserID() : String
+ getEmail() : String
+ setTelNo(telNo : int) : int
+ setUserID(userID : String) : void
+ setEmail(email : String) : void
+ displayMenu() : String

Login

- FILE_NAME: String

+ cust(scan : Scanner) : Customer
+ staff(scan : Scanner) : Staff
- loadCustomers(): List<Customer>

Payment

- bookingID: String
- amount: double
- timestamp: LocalDateTime

+ Payment(bookingId: String, amount: double)
+ getBookingId(): String
+ getAmount(): double
+ getTimestamp(): LocalDateTime
+ toString() : String

PaymentController

- payments: List<Payment>
- FILE_NAME: String

+ PaymentController()
+ processPayment(bookingId: String, seatCount: int, pricePerSeat: double): Payment
+ viewPayments(): void
- savePayments(): void

SeatLayout

- fileName: String
- rows: int
- cols: int
- seatMap: Map<String, Seat>
- showtime: Showtime

+ SeatLayout()
+ SeatLayout(showtimeL Showtime)
- generateSeats(): void
+ getShowtime(): Showtime
+ bookSeat(seatID: String): boolean
+ bookSeats(seatIds: List<String>): List<String>
+ displayLayout(): void
- saveSeats(): void
- loadSeats(): Map<String, Seat>
+ isSeatAvailable(seatId: String): boolean
+ getUnavailableSeats(requestedSeats: List<String>): List<String>

Staff

- staffID : String

+ Staff(name: String password: String, staffID: int)
+ getStaffID() : int
+ setStaffID(staffID: int) : void
+ displayMenu() : String

BookingController

- FILE_NAME: String
- TICKET_FILE: String
- layoutMap: Map<String, SeatLayout>
- bookings: List<Booking>
- tickets: List<Ticket>
- scanner: Scanner
- showtimeController: ShowtimeController
- paymentController: PaymentController
- currentCustomer: Customer

+ BookingController(showtimeController: ShowtimeController, customer: Customer)
+ addBooking(): void
+ saveBookings(): void
+ viewTickets(): void
+ saveTickets(): void
+ getDetailedHallType(hall: Hall):String

CustomerProfile

- FILE_NAME: String

+ edit(customer: Customer, scanner: Scanner): void
- validateEmail(email: String): boolean
- validateName (name : String) : boolean
- validatePassword(password: String, confirmPassword: String): boolean
- saveChanges(customer: Customer): void
- confirmDeleteAccount(Scanner scanner) : boolean
- deleteAccount(Customer customer) : boolean

Person

- name : String
- password : String

+ Person(name: String, telNo: int, password: String)
+ getName() : String
+ getPassword() : String
+ setName(name : String) : void
+ setPassword(password : String) : void
+ displayMenu() : String

Ticket

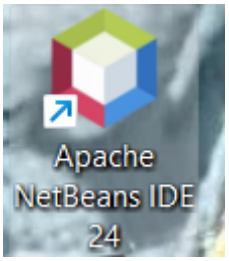
- booking: Booking
- movie: Movie
- hall: Hall
- showtime: Showtime
- seats: List<String>
- totalAmount: double
- customer: Customer

+ Ticket(booking: Booking , showtime: Showtime, seats: List<String>, totalAmount: double, customer: Customer)
+ accessors and mutators
+ toString() : String

4.0 System Workflows

4.1 Main Program

Note : Use **Apache NetBeans IDE 24 to run** our system for the best result. Thank you.



4.1.1 Home

```
|-----------------------------------------------------------|
|                                                           |
|              WELCOME TO CINEMA TICKETING SYSTEM           |
|                                                           |
|-----------------------------------------------------------|
|                          MAIN MENU                        |
|-----------------------------------------------------------|
|                                                           |
|                     [1.] Customer Login                   |
|                     [2.] Staff Login                      |
|                     [3.] Register Customer                |
|                     [0.] Exit                             |
|                                                           |
|-----------------------------------------------------------|

Enter your choice:
```

The Main Menu option include:

[1.] Customer Login

- Allows the customer to log into the system with their ID and password.

[2.] Staff Login

- Allow staff to log into the system with the default **staff ID**: admin and **password**: ******.

[3.] Register Customer

- Allow customers to register for an account with their personal details.

[4.] Exit

- Exit the program.

4.1.2 Staff Login

notes: In the main menu, enter the choice 2 enter into staff login.

```
Enter your choice:2

            |------------------------|
            |      Staff Login       |
            |------------------------|

Enter Staff ID: admin
Enter Password: ******
```

Explanation: Enter the default staff ID and password for staff login.

4.1.2.1 Staff Menu

- After login successfully, the staff menu will be displayed.

```
 ____________________________________________________
|                                                    |
|                    Staff Menu                      |
|____________________________________________________|
|                                                    |
|              [1.] Movie Management                 |
|                                                    |
|              [2.] Showtime Management              |
|                                                    |
|              [3.] Hall Management                  |
|                                                    |
|              [0.] Back                             |
|____________________________________________________|
|                                                    |
|       Choose with [1.], [2.], [3.] & [0.]          |
|____________________________________________________|
Enter your choice:
```

The Staff Menu option include:

[1.] Movie Management

- Allow the staff to add, view, update, delete, search movies and back to the staff menu.

[2.] Showtime Management

- Allow the staff to add, view, delete showtimes and back to the staff menu.

[3.] Hall Management

- Allow the staff to add, view, delete halls and back to the staff menu.

[0.] Back

- Back to the Main menu.

4.1.3 Movie Management

notes: In the staff menu, enter the choice 1 enter into movie management.

```
Enter your choice: 1

================================
=   Movie Management System    =
================================

[1.] Add Movie
[2.] View Movies
[3.] Update Movie
[4.] Delete Movie
[5.] Search Movie
[0.] Back
Enter your choice:
```

The Movie Management System option include:

[1.] Add Movie

- Allow the staff to add the movies.

[2.] View Movie

- Allow the staff to view the movies.

[3.] Update Movie

- Allow the staff to update the movie's details.

[4.] Delete Movie

- Allow the staff to delete the movies.

[5.] Search Movie

- Allow the staff to search the movie to display more details.

[0.] Back

- Back to the staff menu.

4.1.3.1 Add Movie

notes: In the Movie Management System, enter the choice 1 to add a movie.

```
Enter your choice: 1

============================
=      Add New Movie       =
============================

Generated Movie ID: aa3ac0ec
Enter movie title: A Minecraft Movie
Enter genre: American fantasy adventure comedy
Enter running time: 1H41M
Enter subtitles: English
Enter director: Jared Hess
Enter casts: Steve|Jason Momoa|Jack Black|Emma Myers...
Enter spoken language: English
Enter description: In the film, four misfits are pulled
Enter classification: PG
Movie added successfully with ID: aa3ac0ec
```

Steps:

notes: Movie ID will auto generate without staff input manually.

1. Enter movie title for new movie.
2. Enter genre for new movie.
3. Enter running time for the new movie.
4. Enter subtitles for the new movie.
5. Enter the director name for the movie.
6. Enter casts for the movie.
7. Enter the spoken language for the movie.
8. Enter the description for the movie.
9. Enter the classification for the movie.

Output: After inputting all the information, a successful message will appear to notify the staff. And will automatically direct back to the Movie Management System.

4.1.3.2 View Movie

notes: In the Movie Management System, enter the choice 2 to view a movie.

```
======================================
=             Movie List             =
======================================


--------------------------------------------------------------------------
|                          MOVIE INFORMATION                             |
--------------------------------------------------------------------------
| ID              : 3ef75ca3                                             |
| Title           : NaZha2                                               |
| Genre           : Animation                                            |
| Running Time    : 2H30M                                                |
| Subtitles       : Chinese                                              |
| Director        : JiaoZi                                               |
| Casts           : Nazha,AoBing                                         |
| Spoken Language : Mandarin                                             |
| Description     : Two kid...                                           |
| Classification  : PG13                                                 |
--------------------------------------------------------------------------


--------------------------------------------------------------------------
|                          MOVIE INFORMATION                             |
--------------------------------------------------------------------------
| ID              : 5f3ad52f                                             |
| Title           : TheWinning                                           |
| Genre           : Concert                                              |
| Running Time    : 2H4M                                                 |
| Subtitles       : English                                              |
| Director        : LeeYeJi                                              |
| Casts           : IU                                                   |
| Spoken Language : Korean                                               |
| Description     :                                                      |
| Classification  : PG123                                                |
--------------------------------------------------------------------------
```

Output: All the Movie list will display with their movie ID and title. After displaying it will automatically direct back to the Movie Management System.

4.1.3.3 Update Movie

notes: In the Movie Management System, enter the choice 3 to update a movie details.

```
Enter your choice: 3

============================
=       Update Movie       =
============================

Enter the movie ID to update: aa3ac0ec
Editing movie: A Minecraft Movie
Enter new title (press Enter to keep current):
Enter new genre (press Enter to keep current):
Enter new running time (press Enter to keep current):
Enter new subtitles (press Enter to keep current):
Enter new director (press Enter to keep current):
Enter new casts (press Enter to keep current):
Enter new language (press Enter to keep current):
Enter new description (press Enter to keep current):
Enter new classification (press Enter to keep current):
Movie updated successfully!
```

Steps:

1. Enter movie ID to update the movie.
   output: The movie name under editing displayed to staff.
   notes: If the staff want to keep current information, just press enter.
2. Enter a new title for the movie.
3. Enter a new genre for the movie.
4. Enter a new running time for the movie.
5. Enter new subtitles for the movie.
6. Enter a new director for the movie.
7. Enter a new cast for the movie.
8. Enter a new language for the movie.
9. Enter a new description for the movie.
10. Enter a new classification for the movie.

Output: After updating all the details of the movie, a successful message will appear to notify the staff. And will automatically direct back to the Movie Management System.

4.1.3.4 Delete Movie

notes: In the Movie Management System, enter the choice 4 to delete a movie.

```
Enter your choice: 4

=========================
=      Delete Movie      =
=========================


=====================================
=             Movie List            =
=====================================

ID: 69c8af22 | Title: NaZha2
ID: 9f7218d4 | Title: TheWinning
ID: c05e0383 | Title: Avengers: Endgame
ID: aa3ac0ec | Title: A Minecraft Movie
ID: de59b8fd | Title: test

Enter the movie ID to delete: de59b8fd
Movie deleted successfully!
```

output: Display the movie list with movie ID and title.

Steps:

1. Enter the Movie ID to delete the movie.

   output: After successfully deleting a movie, a successful message will appear to notify the staff. And will automatically direct back to the Movie Management System.

4.1.3.5 Search Movie details

notes: In the Movie Management System, enter the choice 5 to search for a movie.

```
Enter the movie ID: a9cc216b
= Movie Found =
-------------------------------------------------------------------------
|                          MOVIE INFORMATION                            |
-------------------------------------------------------------------------
| ID              : a9cc216b                                            |
| Title           : A Minecraft Movie                                   |
| Genre           : American fantasy adventure comedy                   |
| Running Time    : 1H31M                                               |
| Subtitles       : English                                             |
| Director        : Jared Hess                                          |
| Casts           : Steven, Emma...                                     |
| Spoken Language : English                                             |
| Description     : In the film, four ...                               |
| Classification  : PG                                                  |
-------------------------------------------------------------------------
```

Steps:

1. Enter the Movie ID to search the movie.

   output: After entering the correct movie Id to search for a movie, the movie details will display to the staff. And will automatically direct back to the Movie Management System.

4.1.4 Showtime Management

notes: In the staff menu, enter the choice 2 enter into showtime management.

```
Enter your choice: 2

============================
=   Showtime Management    =
============================

[1.] Add Showtime
[2.] View Showtimes
[3.] Delete Showtime
[0.] Back
Choice:
```

The Showtime Management System option include:

[1.] Add Showtime

- Allow the staff to add the showtimes.

[2.] View Showtime

- Allow the staff to view the showtimes.

[3.] Delete Showtime

- Allow the staff to delete the showtimes.

[0.] Back

- Back to the staff menu.

4.1.4.1 Add Showtime

notes: In the Showtime Management System, enter the choice 1 to add a showtime.

```
========================
=   Add New Showtime   =
========================


========================
=       Hall List      =
========================

Hall Type = NormalHall, id = 263, hallName = Hall 1, Base Price = 38.00, TotalPrice = 34.20, HasReclinerSeats = true
Hall Type = ThreeDHall, id = 692, hallName = Hall 2, Base Price = 168.00, TotalPrice = 204.80, Provides3DGlasses = true
Hall Type = IMAXHall, id = 3b5, hallName = Hall 3, Base Price = 388.00, TotalPrice = 465.60, SoundSystemBrand = Dolby Atmos
Hall Type = NormalHall, id = d66, hallName = Hall 5, Base Price = 38.00, TotalPrice = 34.20, HasReclinerSeats = false
Enter Hall ID:
```

output: The system will display the Hall list with hall ID and name to the staff.

Steps:

1. Enter the Hall Id to choose which hall that wants to add a showtime.

```
======================================
=             Movie List             =
======================================


--------------------------------------------------------------------------
|                          MOVIE INFORMATION                             |
--------------------------------------------------------------------------
| ID              : 3ef75ca3                                             |
| Title           : NaZha2                                               |
| Genre           : Animation                                            |
| Running Time    : 2H30M                                                |
| Subtitles       : Chinese                                              |
| Director        : JiaoZi                                               |
| Casts           : Nazha,AoBing                                         |
| Spoken Language : Mandarin                                             |
| Description     : Two kid...                                           |
| Classification  : PG13                                                 |
--------------------------------------------------------------------------
```

output: After input the hall id, the system will display the movie List with the movie id and name.

2. Enter the movie id to choose which movie that wants to add a showtime.

```
Enter Movie ID: aa3ac0ec

Available Dates:
1. 2025-04-27
2. 2025-04-28
3. 2025-04-29
4. 2025-04-30
Select date (1-4): 4
Enter showtimes for 2025-04-30 (e.g., 13:30). Type 'done' to finish:
Time: 13:30
Time: 15:50
Time: 17:40
Time: done
3 showtime added successfully.
```

output: After input the movie id, the system will display the available dates.

3. Enter 1, 2 , 3 , 4 for the dates .

output: After selecting the date, ask the staff to enter the time for the showtime, meaningful example of time and type done to finish adding the time. Notify the staff with a successful message and it will automatically direct back to the Showtime Management System.

4.1.4.2 View Showtime

notes: In the Showtime Management System, enter the choice 2 to view a showtime.

```
================================================================================
=                              Showtime List                                   =
================================================================================


=================
=  2025-05-03   =
=================
====================================
Showtime ID   : c08fd7
Date & Time   : 2025-05-03 13:30
Hall Type     : IMAX
Hall Name     : Hall 3
Movie Title   : TheWinning
====================================

=================
=  2025-05-05   =
=================
====================================
Showtime ID   : fa1ca1
Date & Time   : 2025-05-05 13:30
Hall Type     : Normal
Hall Name     : Hall 1
Movie Title   : A Minecraft Movie
====================================
====================================
Showtime ID   : 0c9ec3
Date & Time   : 2025-05-05 13:30
Hall Type     : 3D
Hall Name     : Hall 2
Movie Title   : Avengers:Endgame
====================================
```

Output: All the Showtime list will display with their showtime ID, time, title and hall. After displaying it will automatically direct back to the Showtime Management System.

4.1.4.3 Delete Showtime

notes: In the Showtime Management System, enter the choice 3 to delete a showtime.

```
==============================================
=               Delete Showtime              =
==============================================


================================================================================
=                                Showtime List                                 =
================================================================================


=================
=  2025-05-03   =
=================
====================================
Showtime ID   : c08fd7
Date & Time   : 2025-05-03 13:30
Hall Type     : IMAX
Hall Name     : Hall 3
Movie Title   : TheWinning
====================================

=================
=  2025-05-05   =
=================
====================================
Showtime ID   : falca1
Date & Time   : 2025-05-05 13:30
Hall Type     : Normal
Hall Name     : Hall 1
Movie Title   : A Minecraft Movie
====================================
====================================
Showtime ID   : 0c9ec3
Date & Time   : 2025-05-05 13:30
Hall Type     : 3D
Hall Name     : Hall 2
Movie Title   : Avengers:Endgame
====================================
====================================
Showtime ID   : aff802
Date & Time   : 2025-05-05 13:30
Hall Type     : Normal
Hall Name     : Hall 5
Movie Title   : test
====================================

Enter Showtime ID to delete: aff802
Showtime with ID aff802 has been deleted.
```

Can be improve

output: Display the showtime list with showtime ID, time, title and hall.

Steps:

1. Enter the Showtime ID to delete the showtime.

   output: After successfully deleting a showtime, a successful message will appear to notify the staff. And will automatically direct back to the Showtime Management System.

4.1.5 Hall Management

notes: In the staff menu, enter the choice 3 enter into hall management.

```
Enter your choice: 3

============================
=      Hall Management     =
============================

[1.] Add Hall
[2.] View Halls
[3.] Delete Hall
[0.] Back
Choice:
```

The Hall Management System option include:

[1.] Add Hall

- Allow the staff to add the hall.

[2.] View Hall

- Allow the staff to view the hall.

[3.] Delete Hall

- Allow the staff to delete the hall.

[0.] Back

- Back to the staff menu.

4.1.5.1 Add Hall

notes: In the Hall Management System, enter the choice 1 to add a hall.

```
Choice: 1

============================
=        Add New Hall      =
============================

Generated Hall ID: 6c5
Enter hall name: test
Select Hall Type:
[1] Normal
[2] 3D
[3] IMAX
Choice: 3
Enter sound system brand (e.g., Dolby Atmos): No
Hall added successfully: Hall Type = IMAXHall, id = 6c5, hallName = test, Base Price = 388.00, TotalPrice = 465.60, SoundSystemBrand = No
```

output: The system will display the auto generated Hall ID.

Steps:

1. Enter the Hall name.
2. Choose the HallType

output: After entering the hall name and hall type, a successful message to notify the staff. And will automatically direct back to the Hall Management System.

4.1.5.2 View Hall

notes: In the Hall Management System, enter the choice 2 to view a hall.

```
Choice: 2

========================
=       Hall List      =
========================

Hall Type = NormalHall, id = 263, hallName = Hall 1, Base Price = 38.00, TotalPrice = 34.20, HasReclinerSeats = true
Hall Type = ThreeDHall, id = 692, hallName = Hall 2, Base Price = 168.00, TotalPrice = 204.80, Provides3DGlasses = true
Hall Type = IMAXHall, id = 3b5, hallName = Hall 3, Base Price = 388.00, TotalPrice = 465.60, SoundSystemBrand = Dolby Atmos
Hall Type = NormalHall, id = d66, hallName = Hall 5, Base Price = 38.00, TotalPrice = 34.20, HasReclinerSeats = false
Hall Type = IMAXHall, id = 6c5, hallName = test, Base Price = 388.00, TotalPrice = 465.60, SoundSystemBrand = No
```

Output: All the hall lists will display with their hall id and name. After displaying it will automatically direct back to the Hall Management System.

4.1.5.3 Delete Hall

notes: In the Hall Management System, enter the choice 3 to delete a hall.

```
==============================
=         Delete Hall        =
==============================


========================
=       Hall List      =
========================

Hall Type = NormalHall, id = 263, hallName = Hall 1, Base Price = 38.00, TotalPrice = 34.20, HasReclinerSeats = true
Hall Type = ThreeDHall, id = 692, hallName = Hall 2, Base Price = 168.00, TotalPrice = 204.80, Provides3DGlasses = true
Hall Type = IMAXHall, id = 3b5, hallName = Hall 3, Base Price = 388.00, TotalPrice = 465.60, SoundSystemBrand = Dolby Atmos
Hall Type = NormalHall, id = d66, hallName = Hall 5, Base Price = 38.00, TotalPrice = 34.20, HasReclinerSeats = false
Hall Type = IMAXHall, id = 6c5, hallName = test, Base Price = 388.00, TotalPrice = 465.60, SoundSystemBrand = No
Enter the hall ID to delete: 6c5
Hall deleted successfully!
```

output: Display the hall list with hall ID and name for confirmation.

Steps:

1. Enter the hall ID to delete the hall.

   output: After successfully deleting a hall, a successful message will appear to notify the staff. And will automatically direct back to the Hall Management System.

4.1.6 Register Customer

notes: In the main menu, enter the choice 3 to register a customer account.

```
Enter your choice:3
          |------------------------|
          |   Register Customer    |
          |------------------------|
Enter your name: test
Enter your username: test
Enter your telNo: 0123456789
Enter your email: test@gmail.com
Enter your password (must be at least 8 characters long including numbers(0-9) and alphabets): ab123456
Reenter your password: ab123456
```

Steps:

1. Enter customer name.
2. Enter customer username.
3. Enter customer handphone number.
4. Enter customer email.
5. Enter password.
6. Re-enter password.

```
Please check your personal details:
=====================================
userID: test
telNo : 123456789
email : test@gmail.com
=====================================

Is this information correct? (1. yes, 2. no):
```

Output: After correctly inputting all the information, a confirmation message will appear to notify the customer. The customer needs to make a choice on 1.yes or 2.no.
If [1.yes]

Output: It will appear a successful message and back to the main menu.

```
Is this information correct? (1. yes, 2. no): 1
Registration successful!
```

else
Output: It will appear a re-enter message and back to the main menu.

```
Is this information correct? (1. yes, 2. no): 2
Please re-enter your details.

|-----------------------------------------------------------|
|                                                           |
|              WELCOME TO CINEMA TICKETING SYSTEM           |
|                                                           |
|-----------------------------------------------------------|
|                          MAIN MENU                        |
|-----------------------------------------------------------|
|                                                           |
|                    [1.] Customer Login                    |
|                    [2.] Staff Login                       |
|                    [3.] Register Customer                 |
|                    [0.] Exit                              |
|                                                           |
|-----------------------------------------------------------|

Enter your choice:
```

4.1.7 Customer Login

notes: In the main menu, enter the choice 1 enter into customer login.

```
Enter your choice:1

            |------------------------|
            |     Customer Login     |
            |------------------------|

Enter Customer ID: test
Enter Password: ab123456
Login successful! Welcome, test
```

Explanation: Enter the registered customer ID and password.

Output: If the customer successfully login, will display a login successful message.

4.1.7.1 Customer Menu

- After customer login successfully, the customer menu will be displayed.

```
 ____________________________________________________
|                                                    |
|                   Customer Menu                    |
|____________________________________________________|
|                                                    |
|                [1.] Movie Listing                  |
|                                                    |
|                [2.] View Showtime                  |
|                                                    |
|                [3.] Make Booking                   |
|                                                    |
|                [4.] View My Tickets                |
|                                                    |
|                [5.] Edit Profile                   |
|                                                    |
|                [0.] Back                           |
|____________________________________________________|
|                                                    |
|   Choose with [1.], [2.], [3.], [4.],[5.] & [0.]   |
|____________________________________________________|
Enter your choice:
```

The Customer Menu option include:

[1.] Movie Listing

- Allow the customer to view the available movie.

[2.] View Showtime

- Allow the customer to view the showtimes.

[3.] Make Booking

- Allow the customer to make a booking, make payment and get the tickets.

[4.] View My Tickets

- Allow the customer to view their ticket after successfully booking a ticket.

[5.] Edit Profile

- Allow the customer to modify their personal details.

[0.] Back

- Back to the Main menu.

4.1.7.2 Movie Listing

notes: In the Customer Menu, enter the choice 1 to view the movie list.

```
Enter your choice: 1

======================================
=             Movie List             =
======================================

ID: 3ef75ca3 | Title: NaZha2
ID: 5f3ad52f | Title: TheWinning
ID: f7b0b8d4 | Title: Avengers:Endgame
ID: a9cc216b | Title: A Minecraft Movie
```

Output: All the available Movie lists will display with their movie ID and title. After displaying it will automatically direct back to the Customer Menu.

4.1.7.3 View showtimes

notes: In the Customer Menu, enter the choice 2 to view the showtimes.

```
Enter your choice:  2

================================================================================
=                                Showtime List                                 =
================================================================================


=================
=  2025-05-03   =
=================
====================================
Showtime ID   : c08fd7
Date & Time   : 2025-05-03 13:30
Hall Type     : IMAX
Hall Name     : Hall 3
Movie Title   : TheWinning
====================================

=================
=  2025-05-05   =
=================
====================================
Showtime ID   : fa1ca1
Date & Time   : 2025-05-05 13:30
Hall Type     : Normal
Hall Name     : Hall 1
Movie Title   : A Minecraft Movie
====================================
====================================
Showtime ID   : 0c9ec3
Date & Time   : 2025-05-05 13:30
Hall Type     : 3D
Hall Name     : Hall 2
Movie Title   : Avengers:Endgame
====================================
```

output: Display all the available showtimes to customers. And will automatically direct back to the Customer Menu.

4.1.7.4 Edit Profile

notes: In the Customer Menu, enter the choice 5 to Modify the personal information.

```
============================
=       Edit Profile       =
============================

Current Profile Information
---------------------------
Name   : test
User ID: test
Email  : test@gmail.com
Phone  : 123456

What would you like to edit?
[1.] Name
[2.] Email
[3.] Phone Number
[4.] Password
[5.] Delete Account
[0.] Back
Enter your choice:
```

output: The system will display the customer's personal information.

The Edit profile option include:

[1.] Edit name

- Allow the customer to edit their name.

[2.] Edit email address

- Allow the customer to edit their email.

[3.] Edit phone number

- Allow the customer to edit their phone number.

[4.] Edit password

- Allow the customer to edit their password.

[0.] Back

- Back to the customer menu.

4.1.7.4.1 Edit name

```
What would you like to edit?
[1.] Name
[2.] Email
[3.] Phone Number
[4.] Password
[0.] Back
Enter your choice: 1
Enter new name: Tester
Name updated successfully!
Profile changes saved successfully!
```

Steps:

1. Enter the choice 1 to edit the name.
2. Enter the new name.

Output: A successful message will be displayed to the customer. It will direct to the customer menu after success.

4.1.7.4.2 Edit Email

```
============================
=       Edit Profile       =
============================

Current Profile Information
---------------------------
Name   : test
User ID: test
Email  : test@gmail.com
Phone  : 123456

What would you like to edit?
[1.] Name
[2.] Email
[3.] Phone Number
[4.] Password
[5.] Delete Account
[0.] Back
Enter your choice: 2
Enter new email: Tester@gmail.com
Email updated successfully!
Profile changes saved successfully!
```

Steps:

1. Enter the choice 2 to edit the email.
2. Enter the new email.

Output: A successful message will be displayed to the customer. It will direct to the customer menu after success.

4.1.7.4.3 Edit phone number

```
============================
=       Edit Profile       =
============================

Current Profile Information
---------------------------
Name   : test
User ID: test
Email  : Tester@gmail.com
Phone  : 123456

What would you like to edit?
[1.] Name
[2.] Email
[3.] Phone Number
[4.] Password
[5.] Delete Account
[0.] Back
Enter your choice: 3
Enter new phone number: 188888888
Phone number updated successfully!
Profile changes saved successfully!
```

Steps:

1. Enter the choice 3 to edit the phone number.
2. Enter the new phone number.

Output: A successful message will be displayed to the customer. It will direct to the customer menu after success

4.1.7.4.4 Edit password

```
============================
=       Edit Profile       =
============================

Current Profile Information
---------------------------
Name   : test
User ID: test
Email  : Tester@gmail.com
Phone  : 188888888

What would you like to edit?
[1.] Name
[2.] Email
[3.] Phone Number
[4.] Password
[5.] Delete Account
[0.] Back
Enter your choice: 4
Enter current password: ab123456
Enter new password (must be at least 8 characters long including numbers(0-9) and alphabets): test123456
Confirm new password: test123456
Password updated successfully!
Profile changes saved successfully!
```

Steps:

1. Enter the choice 4 to edit the password.
2. Enter the old password.
3. Enter the new password.
4. Re-enter the new password.

Output: A successful message will be displayed to the customer. It will direct to the customer menu after success

4.1.7.4.5 Delete Account

```
============================
=       Edit Profile       =
============================

Current Profile Information
---------------------------
Name   : test
User ID: test
Email  : Tester@gmail.com
Phone  : 188888888

What would you like to edit?
[1.] Name
[2.] Email
[3.] Phone Number
[4.] Password
[5.] Delete Account
[0.] Back
Enter your choice: 5

=======================  WARNING: ACCOUNT DELETEION  =======================
                    You are about tp delete this account.
                 This action CANNOT be reverseble or undone
     All your data including booking history will be permenantly deleted
==============================================================================
Are you sure you want to delete your account? [1.]Yes [2.]No
Enter your choice:  1
Account deleted successfully!
Your account no longer exists. Returning to main menu.
```

Steps:

1. Enter choice 5 to delete the account

   Output: A warning message will be displayed to confirm the customer's decision

   note: the customer can choose to enter 2 to cancel or return to the customer menu if they changed their mind.
2. Enter choice 1 to initiate the cancellation process.

Output: A successful message will be displayed to the customer. It will direct the user to the main menu after success.

4.1.8 Booking Management

notes: In the customer menu, enter the choice 3 to make a booking ticket.

```
Enter your choice:  3

================================================================================
=                                Showtime List                                 =
================================================================================


=================
=  2025-05-03   =
=================
====================================
Showtime ID    : c08fd7
Date & Time    : 2025-05-03 13:30
Hall Type      : IMAX
Hall Name      : Hall 3
Movie Title    : TheWinning
====================================
```

output: The system will display the showtime list to the customer.

Steps:

1. Enter the preferred showtime id.

```
Enter Showtime ID to book: 8e76ab
                SCREEN
                -------
A1 [ ]  A2 [ ]  A3 [ ]  A4 [ ]  A5 [ ]
B1 [ ]  B2 [ ]  B3 [ ]  B4 [ ]  B5 [ ]
C1 [ ]  C2 [ ]  C3 [ ]  C4 [ ]  C5 [ ]
D1 [ ]  D2 [ ]  D3 [ ]  D4 [ ]  D5 [ ]
E1 [ ]  E2 [ ]  E3 [ ]  E4 [ ]  E5 [ ]
Enter Seat to book (e.g: A1,A2,A3):
```

output: The available seat layout will be displayed to the customer to allow customers to choose their preferred seat.

Steps:

2. Enter the seat. The same seat cannot be book.

```
Enter Showtime ID to book (or 0 to cancel): c08f

                SCREEN
                -------
A1 [ ]  A2 [ ]  A3 [ ]  A4 [ ]  A5 [ ]
B1 [ ]  B2 [ ]  B3 [ ]  B4 [ ]  B5 [ ]
C1 [ ]  C2 [ ]  C3 [ ]  C4 [ ]  C5 [ ]
D1 [ ]  D2 [ ]  D3 [ ]  D4 [ ]  D5 [ ]
E1 [ ]  E2 [ ]  E3 [ ]  E4 [ ]  E5 [ ]

Enter Seat to book (e.g: A1,A2,A3): a1,a2
========================================
=           Booking Summary            =
========================================
Movie        : TheWinning
Hall         : IMAX Hall (Dolby Atmos)
Date & Time : 2025-05-03 13:30
Seats        : A1, A2
Price/Seat   : RM465.60
Total Amount: RM931.20
Proceed to payment? (yes/no):
```

output: After entering the seat, the system will display the booking Id and the seats chosen by the customer and with the total price seat (e.g. RM465.60 per seat). And ask if the customer wants to proceed to payment or not.

Steps:

3. Enter yes.

```
Proceed to payment? (yes/no): yes

==================================
=       Payment Successful       =
==================================
Payment ID: P-7fdf71ee
Amount    : RM931.20
Paid On   : 2025-05-02 16:47:08

============ Ticket ============
Ticket ID   : T-98244d0b
Booking ID  : B-c502674d
Movie       : TheWinning
Hall Name   : Hall 3
Hall Type   : IMAX Hall (Dolby Atmos)
Date & Time : 2025-05-03 13:30
Seats       : A1, A2
Total       : RM931.20
Customer    : bob
================================
```

output: After entering yes, a payment successful notification will display to the user and the ticket of the booking details. It will direct back to the Customer Menu after success.

4.1.8.1 Seat Unavailable

explanation: After the customer finished booking, the seat became unavailable.

```
Enter Showtime ID to book: 8e76ab
                SCREEN
                -------
A1 [X]  A2 [ ]  A3 [ ]  A4 [ ]  A5 [ ]
B1 [ ]  B2 [X]  B3 [ ]  B4 [ ]  B5 [ ]
C1 [X]  C2 [ ]  C3 [ ]  C4 [ ]  C5 [ ]
D1 [ ]  D2 [ ]  D3 [ ]  D4 [ ]  D5 [ ]
E1 [ ]  E2 [ ]  E3 [ ]  E4 [ ]  E5 [X]
Enter Seat to book (e.g: A1,A2,A3):
Unavailable Seats:
```

4.1.8.2 View My Tickets

notes: In the Customer Menu, enter the choice 4 to view my tickets.

```
Enter your choice:  4

================================
=         Your Tickets         =
================================


============ Ticket ============
Ticket ID   : T-60d91fbd
Booking ID  : B-783781b9
Movie       : Avengers:Endgame
Hall Name   : Hall 2
Hall Type   : 3D Hall (with glasses)
Date & Time : 2025-05-05 13:30
Seats       : A1, A2
Total       : RM409.60
Customer    : bob
================================


============ Ticket ============
Ticket ID   : T-cabbc5c8
Booking ID  : B-c4804e95
Movie       : Avengers:Endgame
Hall Name   : Hall 2
Hall Type   : 3D Hall (with glasses)
Date & Time : 2025-05-05 13:30
Seats       : A3
Total       : RM204.80
Customer    : bob
================================
```

Output: The ticket booked by the customer will display to the customer, after display will direct back to the customer menu.

5.0 Conclusion

In conclusion, the Cinema Ticketing System for managing the ticketing of movies provides a robust and efficient solution for handling the complexities of movie booking and user accounts. By integrating various modules that address each aspect of movie management, booking management, and user management, the system improves operational efficiency and reduces the incidence of errors. The user management system improves the system by ensuring that access is appropriately controlled based on different characteristics. Overall, this CTS provides valuable insights that empower users to make better decisions for obtaining satisfaction in entertainment.

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